ANCHOR · v0 · THIS SHEET DOES NOT CHANGE

Small edits. *Different worlds.*

How far can a safety specification drift while every revision looks reasonable?

Luiza Corpaci
TU Wien · Apart Research, Secure Program Synthesis · CAISH MARS V
with Malhar Alpesh Patel · *When Specifications Drift*, NeurIPS 2026 VeriCodeGen
corpaci.luiza@gmail.com

THE PROBLEM

$P \models \varphi$
Verification. Does the system satisfy the spec? Provers do this.

$\varphi \approx$ intent
Validation. Does the spec still say what we meant? Nobody does this, and specs get revised all the time.

Guaranteed Safe AI asks for a spec, a world model and a verifier. Changing a definition does not make a proof invalid. Three questions survive every revision: **What was proved? Under which assumptions? Does that still establish what we wanted?**

TWO DISTANCES, MEASURED SEPARATELY

textual How much did the wording change? Token edit distance.

behavioural Which executions became permitted or forbidden? Counted over a fixed, finite universe of executions.

A revision is represented by the set of executions it permits. Against its parent and against v0 it is equivalent · weaker · stronger · mixed or unresolved.

Different meaning is not automatically worse. Separately marked: do newly permitted executions break the budget? Do newly forbidden ones block legitimate work?

WHO ACCEPTS A REVISION

reviewer Eight public executions. A revision is accepted when they are judged exactly as before.

oracle 2,016 executions: 8 public, 8 hidden hand-written, 2,000 sampled from the toy world. Measures both distances.

chain Visitors revise the *latest* accepted revision, not v0. The paper keeps the initial intent; the screen shows what revisions have made of it.

SAME GAP, BIGGER TABLES

patch The patch redefines "authorized user". The proof still passes.

compute "No large run on undeclared chips." Undeclared is invisible by construction.

release "No deployment with a dangerous capability unless mitigated." Evals never run.

model spec Model specs ship with textual diffs. Nobody publishes the behavioural one.

THE INITIAL INTENT · V0 · PRINTED HERE, REVISED THERE

One task has a total allowance of 10 compute credits. All computation initiated for that task, including delegated work, retries and restarts, consumes the same allowance, as the independent meter reads it. No execution may exceed it.

```
used:  metered(main) + sum(metered(descendants)) +
metered(retries) + metered(restarts) +
sum(metered(relabelled))
claim: used <= 10
```

A TRAIL OF REASONABLE-LOOKING EDITS

WORDING	PERMITTED EXECUTIONS
"All computation **initiated for** the task..."	main 6, two workers 3 and 3 → refused (12 > 10)
↓	↓
"All computation **performed by** the task..."	workers' spend is no longer "by the task" → certified at 6
↓	↓
"All computation **recorded against** the task..."	work filed under another name is not "recorded against" → certified
Three small textual steps. Each reads like a clarification.	Two behavioural steps. The proof passes at every stage.

At which edit did *the intent* change?

Illustrative candidates, not established equivalences. The page measures the real ones.

Judge a revision. *Then revise one yourself.*



? **You are the overseer.** Same meaning or changed? Commit, get help, judge again, replay the counterexample. Three rounds, two minutes.

≈ **Revise it.** Up to three token edits that change the wording and not the meaning. The reviewer accepts; the oracle tells you how far you really moved.

↯ **Change the intent.** Get a revision accepted that lets an over-budget execution through.

AI **Let a model revise it.** Scan a model below; it gets the latest revision and the task. Paste its answer back.

https://corpaci.com/fork-this-spec/



Claude
revises the latest, prompt pasted



ChatGPT
revises the latest, prompt pasted



Perplexity
revises the latest, prompt pasted



Change the intent
straight into the adversarial role

FOR THE CONVERSATION

WHAT THE COMPANION PAPER MEASURED

500 algebraic laws · 18 representations · 9 models · a certified implication oracle from the Equational Theories Project.

73.7 → 8.5%
certified recovery, from bracket words down to postfix notation

30.3%
provably weaker on confusable identifiers; 93.7% of those merged variables

71 / 38 / 25%
meaning surviving six translation hops, three models, exploratory

Selecting on validity moved failures from loud to silent without improving survival. Here the reviewer is the selector.

YOU ARE THE OVERSEER

Two rules, original and revision. **Same meaning or changed?** Commit, then get help assigned at random: a second look, an AI critique, or a critique with a witness execution you can replay. Judge again. Then the oracle reveals the truth. Some critiques lie.

Tallied per form of help: drift caught, false alarms, judgments rescued, judgments misled. Does assistance make the decision easier to *verify*, or only easier to believe? **Exact equivalence is transitive:** if every revision had preserved meaning, the chain could not drift.

THE TOY WORLD BEHIND THE DISTANCES

A main agent runs a task with ten credits; it can delegate, workers can delegate further; attempts fail and are retried; a restart gives a fresh counter; a worker can file its work under another task's name; every unit of work is metered and self-reported. Six ways a revision can leave the promise: children for descendants, retries uncounted, relabelled work excluded, restart resets the counter, max for sum, self-report for the meter. The last is tool-call spoofing, which METR found in at least 96 transcripts of the July 2026 OpenAI evaluation.

ASK ME ABOUT

1. Iterated revision with a reviewer is a refinement process. Does it converge, and to whose intent?
2. Which clauses should never be allowed to weaken, and who decides?
3. Can a revision step be typed as a lens whose backward map carries the preservation certificate?

STICK A DOT WHERE YOUR ACCEPTED REVISION LANDED

0	1–10	11–50	51–200	201–1,000	> 1,000
intent intact	executions flipped				of 2,016

Behavioural distance from v0 (the page shows it after the reviewer accepts)

Behavioural distance from the revision you started from

Orange dot if the oracle says over-budget executions are now certified; blue if legitimate work is now refused; green if the initial intent is intact. Distances are over the 2,016 executions described above.

The paper preserves the initial intent. The screen reveals what successive revisions have made of it. Every accepted revision is kept with its author or model, its parent, both distances and a witness execution. The chain and the hidden executions are published after the event. · Boehm 1984 · Dalrymple et al. 2024 · METR, Aug 2026 · Shovelain & Skalse 2022 · Videla 2024 · Patel & Corpaci 2026

Bring your model.
corpaci.luiza@gmail.com